VIAVI UE Emulator: Uplink CoMP Reception in PRB Overlapping Regions

Technical Troubleshooting Guide, Multi-Cell Architecture Diagrams, and 5G NR FR1 / LTE PHY & RRC Verification Reference (TM500 / E500)

> **Overview:** When testing **Uplink Coordinated Multi-Point (UL CoMP) Reception** or 5G NR **Multi-TRP UL Joint Reception** using a VIAVI UE Emulator (TM500 or E500), failures in a **Physical Resource Block (PRB) overlapping region** typically stem from timing synchronization mismatches, DMRS scrambling/port collisions, uplink power control imbalances, or O-RAN 7.2x fronthaul alignment errors.

1. Common Failure Symptoms

In a PRB-overlapping UL CoMP scenario—where two UEs or spatial layers share identical time-frequency PRBs across cooperating cells/TRPs—the UE emulator and base station (gNB/eNB) typically exhibit:

- **High PUSCH BLER / CRC Failures:** Cooperating reception points fail to separate or coherently combine overlapping uplink streams.
- **HARQ Retransmission Loops:** Excessive NACKs occur even when simulated path loss and SNR appear favorable.
- **UL Throughput Degradation:** The scheduler falls back to conservative Modulation and Coding Schemes (MCS) or drops out of CoMP Joint Reception (JR) into non-CoMP single-TRP reception.
- **DMRS Channel Estimation Errors:** Poor SINR reporting on uplink DMRS causes inaccurate Interference Rejection Combining (IRC) or joint MMSE equalization at the BBU/O-DU.

2. Primary Root Causes & Configuration Pitfalls

- **Timing Advance (TA) & Propagation Delay Mismatch:** In UL CoMP Joint Reception, signals arrive at two TRPs with different propagation delays. If the VIAVI channel simulator applies asymmetrical delays between the UE and cooperating cells that exceed the Cyclic Prefix (CP) window, Inter-Symbol Interference (ISI) corrupts the overlapping PRBs.
- **DMRS Port & Scrambling ID (`n_SCID`) Collisions:** When multiple UEs share overlapping PRBs in UL MU-MIMO / CoMP, the receiver relies on orthogonal DMRS ports or shared virtual cell scrambling identities. If the VIAVI UE script uses hardcoded DMRS initialization parameters rather than dynamically following RRC reconfiguration, the receiver cannot decouple the streams.
- **Uplink Power Control (P_0 and α) Imbalance:** A UE transmitting at high power to reach a distant cooperating TRP can drown out a nearby UE sharing the same PRBs ("near-far" effect) if fractional path-loss compensation (α) or closed-loop TPC commands per TRP/SRS resource set are misconfigured.
- **Non-Contiguous PRB / Fronthaul (O-RAN 7.2x) Support:** Over eCPRI fronthaul (Option 7-2x), overlapping or fragmented PRB allocations require proper fronthaul licensing and strict eCPRI Section Type 1 alignment (`startPrbc` / `numPrbc`) so PRBs are not truncated before reaching the O-DU equalizer.
- **SRS Configuration Mismatch:** Accurate UL CoMP scheduling relies on fresh Channel State Information (CSI) from UE Sounding Reference Signals (SRS). Mismatched SRS bandwidth, comb offset, or periodicity leads the scheduler to co-schedule spatially correlated UEs on overlapping PRBs.

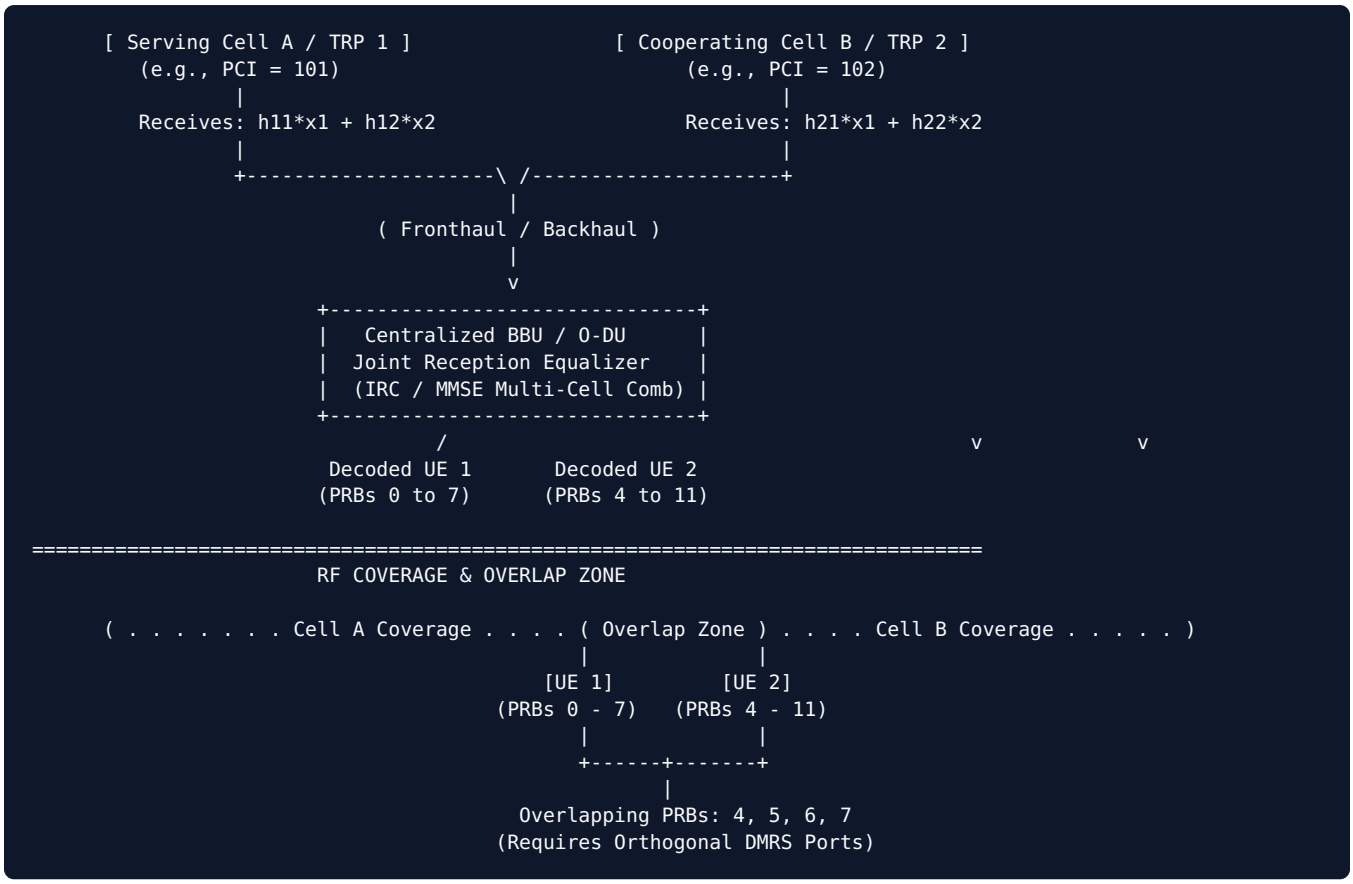
3. Troubleshooting & Verification Checklist

- **Verify Channel Model Delays:** Check the VIAVI embedded fading/channel emulator settings between each UE and TRP. Ensure differential delay stays well within the normal CP duration (< 4.7 µs for 15 kHz SCS; < 2.34 µs for 30 kHz SCS).
- **Inspect RRC & DCI Logs on TM500/E500:** Confirm the UE parses DCI Format 0_1 / 0_2 fields for *SRS Resource Indicator (SRI)* and *Antenna Ports / DMRS sequence initialization*, and verify virtual UEs in the overlap zone use orthogonal DMRS ports or shared virtual scrambling IDs.
- **Check Uplink Power Headroom (PHR) & TPC:** Review TM500/E500 PHY logs to ensure UE transmit power tracks gNB TPC commands without hitting `P_CMAX` prematurely.
- **Isolate CoMP Modes:** Validate *Coordinated Scheduling/Beamforming (CS/CB)* with non-overlapping PRBs first before enabling *Joint Reception (JR)* with PRB overlap to separate RF/timing issues from baseband IRC equalizer issues.
- **Verify 3GPP Release Support:** Confirm the TM500/E500 software build supports the specific Release 16/17 Multi-TRP PUSCH repetition or simultaneous multi-panel UL transmission features under test.

4. General Two-Cell UL CoMP Architecture & PRB Grid

In **Uplink CoMP Joint Reception (UL CoMP-JR)**, UE 1 (Cell A) and UE 2 (Cell B) sit in the cell-edge overlap zone and transmit simultaneously on shared PRBs (PRBs 4–7). Both TRPs forward baseband IQ samples to a centralized BBU/O-DU for joint equalization.

4.1 Spatial & Network Topology (Cell-Edge Overlap)



4.2 Frequency-Domain PRB Allocation Grid (1 Slot)

PRB Index	Cell A Allocation (UE 1)	Cell B Allocation (UE 2)	UL CoMP Processing Mode
PRB 0	[UE 1 PUSCH]	(Unused)	Single-Cell / Non-Overlapping
PRB 1	[UE 1 PUSCH]	(Unused)	Single-Cell / Non-Overlapping
PRB 2	[UE 1 PUSCH]	(Unused)	Single-Cell / Non-Overlapping
PRB 3	[UE 1 PUSCH]	(Unused)	Single-Cell / Non-Overlapping
PRB 4	[UE 1 (DMRS Port 0)]	[UE 2 (DMRS Port 1)]	>>> PRB OVERLAPPING REGION <<<
PRB 5	[UE 1 (DMRS Port 0)]	[UE 2 (DMRS Port 1)]	Both TRPs receive UE 1 + UE 2.
PRB 6	[UE 1 (DMRS Port 0)]	[UE 2 (DMRS Port 1)]	BBU separates streams via
PRB 7	[UE 1 (DMRS Port 0)]	[UE 2 (DMRS Port 1)]	Joint MMSE-IRC & DMRS orthogonality
PRB 8	(Unused)	[UE 2 PUSCH]	Single-Cell / Non-Overlapping
PRB 9	(Unused)	[UE 2 PUSCH]	Single-Cell / Non-Overlapping
PRB 10	(Unused)	[UE 2 PUSCH]	Single-Cell / Non-Overlapping
PRB 11	(Unused)	[UE 2 PUSCH]	Single-Cell / Non-Overlapping

5. 5G NR FR1 Uplink CoMP Example (Band n78, 30 kHz SCS)

In 5G NR FR1 (e.g., Band n78 at 3.5 GHz, 30 kHz SCS, 100 MHz = 273 PRBs), UL CoMP with PRB overlap operates as **Cross-Cell UL MU-MIMO** using shared **Virtual Cell IDs**, **Orthogonal DMRS CDM Groups**, and centralized O-DU Joint MMSE-IRC processing.

5.1 NR FR1 Multi-Cell / Multi-TRP Topology



5.2 NR FR1 Frequency-Domain PRB Grid (1 Slot = 14 Symbols, 0.5 ms)

NR PRB Index	Cell 1 Allocation (UE 1)	Cell 2 Allocation (UE 2)	O-DU Receiver Processing
PRB 0 - 49	[UE 1 PUSCH]	(Unused)	Single-TRP Reception (Cell 1)
PRB 50 - 99	[UE 1 PUSCH • DMRS Port: 1000 (CDM Grp 0) • Virtual DMRS ID: 500 • Data Scrambling ID: 500]	[UE 2 PUSCH • DMRS Port: 1002 (CDM Grp 1) • Virtual DMRS ID: 500 • Data Scrambling ID: 500]	>>> PRB OVERLAPPING REGION <<< • Both TRPs forward IQ to O-DU • Joint MMSE-IRC separates UE 1 & 2 • Requires < 2.34 µs delay delta
PRB 100 - 149	(Unused)	[UE 2 PUSCH]	Single-TRP Reception (Cell 2)

5.3 Critical NR FR1 Parameters for VIAVI Verification

- **Virtual Cell ID Alignment (`scramblingID0` / `dataScramblingIdentityPUSCH`):** Both UEs must receive identical higher-layer scrambling IDs (e.g., `500`) via RRC Reconfiguration rather than defaulting to their serving PCIs (101 and 102).
- **DMRS CDM Group Separation (`DCI Format 0_1 / 0_2`):** Assigning Port 1000 (CDM Group 0) to UE 1 and Port 1002 (CDM Group 1) to UE 2 separates reference signals onto even vs. odd subcarriers inside PRBs 50–99.

- **30 kHz SCS Cyclic Prefix Window (~2.34 µs):** Differential propagation delay plus multipath delay spread between UE 1 → TRP 2 and UE 2 → TRP 1 must remain strictly below ~2.3 µs (~700 m path delta).
- **O-RAN 7.2x Fronthaul Section Handling:** Verify that eCPRI Section Type 1 packets align `startPrbc` and `numPrbc` across Cell 1 (PRBs 0–99) and Cell 2 (PRBs 50–149) in the O-DU symbol buffer.

6. VIAVI TM500 / E500 PHY & RRC Counter Troubleshooting Matrix

Use the reference matrix below to map specific VIAVI UE emulator log indicators and baseband alarms to root causes and corrective actions during PRB-overlapping UL CoMP tests.

VIAVI Log / Counter Symptom	Likely Root Cause	Parameter / IE to Inspect	Corrective Action
High PUSCH CRC Fail / BLER isolated strictly to overlapping PRBs (e.g., PRBs 50–99), while non-overlapping PRBs pass at 0% BLER.	DMRS sequence collision or non-orthogonal DMRS port assignment across cooperating cells.	RRC: `DMRS-UplinkConfig` (`scramblingID0`), `PUSCH-Config` (`dataScramblingIdentityPUSCH`); DCI 0_1/0_2 Antenna Ports.	Configure identical Virtual Cell IDs across both cells and assign orthogonal DMRS ports in separate CDM groups (e.g., Port 1000 vs. Port 1002).
PUSCH SINR Degradation & ISI across all overlapping PRBs when secondary TRP path is enabled in channel emulator.	Inter-TRP propagation delay differential exceeds the OFDM Cyclic Prefix (CP) duration.	VIAVI Channel Model Delay Profile; RRC `n-TimingAdvanceOffset`; MAC Timing Advance Command (TAC).	Reduce differential propagation delay between UE → TRP 1 and UE → TRP 2 to < 2.34 µs (30 kHz SCS) or < 4.7 µs (15 kHz SCS).
UE Tx Power Saturated at `P_CMAX` with persistent positive TPC (+1 / +3 dB) commands in DCI logs, or severe Rx power imbalance (> 15 dB).	Uplink open-loop/ closed-loop power control imbalance ("near-far" saturation at O-RU/BBU ADC).	PHY Log: `PUSCH Tx Power`, `PHR`, `TPC Accumulated`; RRC: `p0-NominalWithGrant`, `alpha`.	Tune fractional path-loss compensation (`alpha`) and target `P0` per TRP/ SRS resource set so received PSD at both TRPs is balanced within 10 dB.

VIAVI Log / Counter Symptom	Likely Root Cause	Parameter / IE to Inspect	Corrective Action
Scheduler Drops Multi-TRP / CoMP Mode (reverts to single-TRP scheduling or low MCS ≤ 4).	Stale or misaligned SRS channel sounding preventing accurate spatial correlation estimation.	RRC: `SRS-Config` (`resourceMapping` , `transmissionComb` , `periodicityAndOffset`); DCI SRI field.	Align UE SRS periodicity, bandwidth, and comb offset with the gNB/O-DU SRS measurement window across both cooperating TRPs.
Fronthaul IQ Truncation / U-Plane Drop in O-RAN 7.2x setups during partial PRB overlap.	eCPRI Section Type 1 PRB indexing mismatch or missing non-contiguous PRB support license.	VIAVI Fronthaul / RU Emulator eCPRI Header: `startPrbc` , `numPrbc` , `sectionId` .	Verify eCPRI U-Plane section mapping aligns overlapping PRB indices in the same slot/symbol buffer and confirm non-contiguous PRB capability is enabled.
RRC Reconfiguration Reject / PHY Assert when enabling simultaneous multi-panel or multi-TRP UL transmission.	UE capability mismatch or unsupported 3GPP Rel-16/17 Multi-TRP feature flag in test script.	`UECapabilityInformation` (`pusch-RepTypeIndicator` , `maxNumberSRS-ResourcePerSet`); TM500/E500 build release notes.	Update VIAVI UE capability profile to declare required Multi-TRP UL features and verify firmware compatibility with the target 3GPP release.